

Yu Li

✉ yul@gwu.edu ☎ (+1) 571-259-8668 🌐 Website 🎓 Scholar in LinkedIn

Education

George Washington University	Washington, D.C.
<i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i>	<i>2025.08 – 2029.05</i>
Advisor: Prof. Tian Lan 🔗 ; Work with: Prof. Zhengling Qi 🔗	<i>(expected)</i>
Wuhan University, Hongyi Honor College	Wuhan, China
<i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>	<i>2021.09 – 2025.06</i>

Experience

ByteDance, Tiktok	San Jose, CA
<i>Research Scientist Intern, Recommendation Core</i>	<i>2026.05 – 2026.08</i>
- Owned the ranking stage of TikTok video recommendation, built on a 3B-parameter Large Recommendation Model. - Adapted LLM architecture techniques to the ranking model, including multi-head latent attention, time-aware RoPE, and mixed-query optimization. - Carried the model from offline training to production deployment, with gains on core business metrics.	

Research Highlights

My research develops principled reinforcement learning methods for **LLM Reasoning**, from token-level credit assignment and preference optimization to on-policy self-distillation [\[NeurIPS'26a\]](#)[\[NeurIPS'26b\]](#)[\[P2\]](#)[\[P3\]](#). These methods extend to **Agentic RL** through hierarchical skill acquisition and multi-turn policy optimization [\[P1\]](#)[\[P4\]](#)[\[ACL'26\]](#). Other work covers **Efficient AI** with parameter-efficient fine-tuning and KV cache eviction [\[COLM'26\]](#)[\[AAAI'26\]](#), as well as **Computer Vision** [\[CVPR'26\]](#).

Preprints & Under Review

P=Preprint

- [\[P1\]](#) **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** [🔗](#)
[Yu Li](#), Rui Miao, Zhengling Qi, Tian Lan
- [\[P2\]](#) **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** [🔗](#)
[Yu Li](#), Tian Lan, Zhengling Qi
- [\[P3\]](#) **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** [🔗](#)
[Yu Li](#), Shu Hong, Tian Lan
Under review. ARR May 2026 OA: 3.33/5
- [\[P4\]](#) **VideoGen-Agent: Reinforcing Video Generation Agents** [🔗](#)
 Binxu Li, Haoyi Duan, Yuhui Zhang, Yaohui Zhang, Zihao Lin, Kaituo Feng, Suozhi Huang, Xiangyi Li, [Yu Li](#), Chunyuan Li, Shilong Liu, Mengdi Wang

Selected Publications [\[Full List\]](#)

- [\[NeurIPS'26a\]](#) **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning**
[Yu Li](#), Rui Miao, Tian Lan, Zhengling Qi

[[NeurIPS'26b](#)] **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization**

[Yu Li](#), Tian Lan, Zhengling Qi

[[COLM'26](#)] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**

[Yu Li](#), Binxu Li, Tian Lan

[[ACL'26 Findings](#)] **Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization**

[Yu Li](#), Sizhe Tang, Tian Lan

[[CVPR'26](#)] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**

[Yu Li](#), Yujun Cai, Chi Zhang

[[ICASSP'26 Oral](#)] **KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models**

[Yu Li](#), Da Chang, Xi Xiao

[[ACM MM'26](#)] **Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers**

Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, [Yu Li](#), Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian

[[AAAI'26](#)] **Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT**

Da Chang, Peng Xue, [Yu Li](#), Yongxiang Liu, Pengxiang Xu, Shixun Zhang

Selected Awards & Honors

-
- **ICML 2026 Gold Reviewer**, top reviewers recognition. *2026*
 - **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. *2024*
 - **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. *2023, 2024*
 - **First-Class Scholarship**, Top 5%, ¥3,000, three times. *2022, 2023, 2024*

Academic Services

-
- **Conference Reviewer:** *2027*: AAAI. *2026*: NeurIPS, ICML, ICLR, COLM, AAAI, ACM MM.
 - **Journal Reviewer:** TPAMI, TMLR, ToN, Neurocomputing.

Skills

-
- **Training Frameworks:** PyTorch, verl, Hugging Face Transformers / Accelerate / TRL, Hydra.
 - **Inference & Distributed:** vLLM, Ray, FSDP, NCCL multi-node training, Docker, Linux GPU clusters (SSH, tmux).
 - **Languages:** English (Proficient), Mandarin (Native), Japanese (Basic).